

On Bruun, Revisited

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Abstract—Bruun’s 1978 real-coefficient Fourier factorization is revisited as a complete algebraic transform rather than only a forward reduction scheme. The starting point is the factor tree of $z^N - 1$ into a binomial spine and real conjugate-pair trinomials. An explicit Chinese-remainder merge supplies a two-sided inverse at every internal node. A normalized coordinate $e_\theta = (z - \cos \theta)/\sin \theta$ turns every terminal quadratic quotient into a Euclidean complex line with $e_\theta^2 = -1$. In these coordinates each nontrivial node is a plane rotation followed by a Hadamard map. Consequently, the packed real transform B_N satisfies $B_N^\top B_N = NI$, and direct evaluation in the terminal quotients proves that B_N is exactly the discrete Fourier transform up to an explicit output permutation. The same factorization admits decimation-in-frequency, decimation-in-time, and diagonal phase-packet walks. Its half-bin companion gives an exact odd-frequency transform using real two-plane arithmetic. Finally, a positive-cone lift and two diagonal gauges give conservative transport and equilibrium realizations whose signed projection is exactly the Fourier map. Symbolic examples and binary64 audits are reported separately from the proofs. The resulting theory identifies an exact, condition-one, real arithmetic Fourier factorization and a physically realizable representative of it in the declared ideal linear categories.

Index Terms—Bruun transform, discrete Fourier transform, Chinese remainder theorem, real FFT, cyclotomic factorization, orthogonal factorization, odd-frequency transform, passive linear network.

I. INTRODUCTION

Let $N = 2^L$ with $L \geq 2$ and associate a real vector $x \in \mathbb{R}^N$ with the residue class

$$x(z) = \sum_{n=0}^{N-1} x_n z^n \quad \text{in} \quad R_N = \mathbb{R}[z]/(z^N - 1). \quad (1)$$

The discrete Fourier transform is the simultaneous evaluation of this class at the N roots of unity. Complex linear algebra is therefore a coordinate choice, not a requirement of the polynomial problem: conjugate roots have the real minimal polynomial $z^2 - 2 \cos \theta z + 1$.

Bruun organized these real factors into a sparse recursive reduction [1]. The construction was remarkable for exposing a real coefficient route to the spectrum, but three questions remain mathematically prior to any implementation question. What is the local inverse of a trinomial split? Which coordinates make the factorization metrically regular? How does the native leaf order identify the ordinary Fourier frequencies? Answering only the first question gives an exact polynomial reconstruction; answering all three identifies a stable Fourier operator.

The present investigation began from those questions. It then asked whether alternative factor bases changed the local multiplication geometry; whether blocks of equal-angle nodes possessed a common real representation; which spectral orders preserve subtree locality; and whether the same factor tree can

be traversed upward rather than downward. Those questions lead, respectively, to a normalized quadratic chart, an isoclinic block form, a classification of admissible leaf orders, and a decimation-in-time dual. Two further questions extend the algebra. Replacing roots of $z^N - 1$ by roots of $z^N + 1$ produces an odd-frequency transform, while shearing the intermediate coordinates produces a diagonal phase-packet transform. A final question asks whether the signed linear cells admit a conservative nonnegative or exact equilibrium realization.

The principal results are as follows.

- 1) An explicit sibling merge gives a two-sided Chinese-remainder inverse for every binomial and trinomial node.
- 2) The normalized terminal coordinate e_θ squares to -1 and converts evaluation into the standard real and imaginary Fourier coordinates.
- 3) Every normalized node is $\sqrt{2}$ times an orthogonal map, giving $B_N^\top B_N = NI$ and condition number one.
- 4) A commuting reduction-evaluation diagram proves $P_N B_N = \mathcal{D}_N^{\mathbb{R}}$ exactly; the factorization is not an approximation to the DFT.
- 5) DIF, DIT, and diagonal phase-packet schedules are boundary-permuted factorizations of that same operator.
- 6) The roots of $z^N + 1$ give an exact real two-plane odd-frequency transform with the same orthogonality law.
- 7) A positive-cone intertwining identity gives exact signed projection through arbitrary compositions, and diagonal gauges convert the cells into conservative transports or convex equilibrium maps.

The mathematical development is intentionally ordered by dependency. Section II reconstructs Bruun’s factor tree and the questions that resume its thread. Sections III–V give the inverse, normalization, and exact Fourier identity. Sections VI–X develop the block, ordering, DIT, ODFT, and DIP consequences. Section XI proves physical realizability in an explicit ideal linear category. Exact examples and finite-precision audits follow only after the algebraic results.

II. BRUUN’S REAL FACTOR TREE

Bruun’s construction begins with the observation that a real DFT need not split $z^N - 1$ into complex linear factors. The two real roots are represented by $z - 1$ and $z + 1$; every other conjugate pair is represented by a real quadratic. The sparse route to those factors is a repeated half-angle identity.

A. Binomial spine and trinomial descendants

For every positive integer m define

$$Q_{m,\theta}(z) = z^{2^m} - 2 \cos \theta z^{2^{m-1}} + 1. \quad (2)$$

The coefficient field throughout the exact development is

$$\mathbb{K}_N = \mathbb{Q}(\sqrt{2}, \{\cos(2\pi k/N), \sin(2\pi k/N) : k \in \mathbb{Z}\}) \subset \mathbb{R}. \quad (3)$$

All polynomial quotients in Sections II–V may be taken over $\mathbb{K}_N$.

Lemma II.1 (Half-angle factorization). *For even m ,*

$$Q_{m,\theta} = Q_{m/2,\theta/2} Q_{m/2,\pi-\theta/2}. \quad (4)$$

Proof. Put $v = z^{m/2}$ and $a = 2\cos(\theta/2)$. The right side is $(v^2 - av + 1)(v^2 + av + 1) = (v^2 + 1)^2 - a^2 v^2$. Since $a^2 = 2 + 2\cos\theta$, this equals $v^4 - 2\cos\theta v^2 + 1$. $\square$

The root polynomial first splits along the binomial spine,

$$z^{2m} - 1 = (z^m - 1)(z^m + 1), \quad (5)$$

and $z^m + 1 = Q_{m/2,\pi/2}$. Repeated application of (4) terminates at

$$q_k(z) = z^2 - 2\cos(2\pi k/N)z + 1, \quad 1 \leq k < N/2, \quad (6)$$

together with $z - 1$ and $z + 1$. Each q_k represents the conjugate frequencies k and $N - k$.

B. Forward reduction

If $p = p_- p_+$ with coprime children, the local forward map is simply

$$[f]_p \mapsto ([f]_{p_-}, [f]_{p_+}). \quad (7)$$

Composing (7) down the complete factor tree gives

$$\mathcal{F}_N : R_N \longrightarrow \mathbb{R} \times \mathbb{R} \times \prod_{k=1}^{N/2-1} \mathbb{R}[z]/(q_k). \quad (8)$$

The two scalar components correspond to $z = 1$ and $z = -1$.

C. The resumed questions

The line of inquiry recorded for the present construction began in November 2024 with four concrete problems: reproduce Bruun’s reduction; identify the leaf shuffle; construct an inverse; and determine whether a normalized form could serve as a complete transform. On June 10, 2026 the thread was resumed from the inverse problem. Claude Fable contributed the decisive local Chinese-remainder reconstruction described in Section III. That formula made the factor tree bidirectional and exposed the next obstruction: exact inversion alone does not choose a well-conditioned coordinate system.

The subsequent questions are stated here because they determine the logical shape of the paper. First, can the terminal quotient be given a unit complex coordinate without introducing complex polynomial arithmetic? Second, do alternative balanced bases improve the local factor geometry? Third, which orders preserve the tree’s sparse intervals? Fourth, do downward, upward, and diagonal walks compute the same boundary map? The answers will be a normalized quadratic chart, a covering-map locality principle, an address algebra, and three exact schedules of one Fourier factorization.

TABLE I
DOCUMENTED MATHEMATICAL DEVELOPMENT LEADING TO THE
PRESENT THEORY.

Date	Mathematical event
Nov. 2024	Bruun reduction, order, inverse, and viability posed as questions.
June 10, 2026	Fable CRT inverse; first staged normalized forward and inverse maps.
June 10–11	Equal-angle block and real rotation inquiry.
June 12	Subtree locality, output order, and first DIT construction.
June 13	Faithful monomial-residue reference reconstructed.
June 14–16	Radix-four and Chebyshev alternative-basis study.
June 23	Exact odd-frequency transform defined independently.
July 4–5	Phase-packet shear, DIP, and twisted span walk.
July 5	Positive-cone and field-delay realizations.
Aug. 13	Corrected cone composition and mechanical gauges.

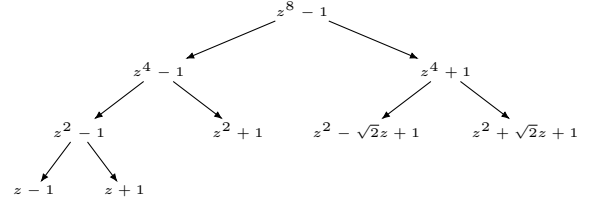


Fig. 1. The $N = 8$ real factor tree. The leaves represent DC, Nyquist, and three conjugate frequency pairs. Every edge is a remainder map; Section III reverses every edge explicitly.

D. Dated reconstruction of the inquiry

Table I records the order in which the questions were resolved. The logical order of the proofs is different: normalization must be established before any schedule can inherit its metric. The June 14–16 Chebyshev work was a return to alternative bases after the first DIT construction, not the origin of the normalized chart present on June 10.

III. EXACT CHINESE-REMAINDER RECONSTRUCTION

The inverse problem is local. Given the two child residues of a parent class, one must reconstruct the unique parent residue without solving a global linear system. The required Bezout coefficient has a closed form for every Bruun sibling pair.

A. Trinomial siblings

Let h be even, $b \neq 0$, and

$$p_- = z^h - bz^{h/2} + 1, \quad p_+ = z^h + bz^{h/2} + 1. \quad (9)$$

Lemma III.1 (Local Bezout coefficient). *In $\mathbb{K}_N[z]/(p_+)$,*

$$p_-^{-1} \equiv \frac{z^{h/2} + b}{2b}. \quad (10)$$

Proof. Modulo p_+ , multiplication by $z^{-h/2}$ gives $z^{-h/2} \equiv -(z^{h/2} + b)$. Also $p_- - p_+ = -2bz^{h/2}$. Hence $p_-(z^{h/2} + b)/(2b) \equiv 1 \pmod{p_+}$. $\square$

Proposition III.2 (Explicit sibling merge). *Let r_- and r_+ be residues modulo p_- and p_+ , respectively. Put*

$$t = \left[\frac{(r_+ - r_-)(z^{h/2} + b)}{2b} \right]_{p_+}, \quad c = [r_- + p_- t]_{p_+ p_-}. \quad (11)$$

Then $c \equiv r_- \pmod{p_-}$ and $c \equiv r_+ \pmod{p_+}$, and c is unique modulo p_-p_+ .

Proof. Choose the canonical polynomial representatives of degrees below h in the two child quotients. The expression in (11) then defines a polynomial lift; changing either lift by its child factor changes c only by a multiple of p_-p_+ . The first congruence is immediate. For the second, use (10) to obtain $p_-t \equiv r_+ - r_- \pmod{p_+}$. Coprimality and the Chinese remainder theorem give uniqueness. $\square$

At a binomial node, $p_- = z^h - 1$ and $p_+ = z^h + 1$. The corresponding merge is

$$t = \frac{r_- - r_+}{2}, \quad c = r_- + (z^h - 1)t. \quad (12)$$

The factors $1/2$ and $1/(2b)$ are the local CRT interpolation measures. Their relation to the uniform inverse Fourier scale is established only after the normalized packed map is identified in Section V.

B. Global inverse and fixed points

Let A_N denote the product algebra on the right side of (8). Define $\mathcal{G}_N : A_N \rightarrow R_N$ by applying (11) and (12) from leaves to root.

Theorem III.3 (Two-sided Bruun reconstruction). *The reduction and reconstruction maps are mutual inverses:*

$$\mathcal{G}_N \mathcal{F}_N = I_{R_N}, \quad \mathcal{F}_N \mathcal{G}_N = I_{A_N}. \quad (13)$$

Proof. At one node, the explicit merge is the inverse of the residue map by the preceding proposition and the binomial identity. Induction on the height of the factor tree proves both compositions in (13). $\square$

Corollary III.4 (Exact fixed-point law). *Every polynomial state is a fixed point of reconstruction after reduction, and every admissible leaf state is a fixed point of reduction after reconstruction. These are identities over $\mathbb{R}$; their structure constants lie in $\mathbb{K}_N$. They are not convergence statements.*

Example III.5 (A symbolic $N = 8$ merge). The siblings $p_- = z^2 - \sqrt{2}z + 1$ and $p_+ = z^2 + \sqrt{2}z + 1$ have $h = 2$ and $b = \sqrt{2}$. For linear residues $r_- = a_0 + a_1z$ and $r_+ = b_0 + b_1z$, the unique quartic parent is

$$c = r_- + p_- \left[\frac{(r_+ - r_-)(z + \sqrt{2})}{2\sqrt{2}} \right]_{p_+}. \quad (14)$$

Substitution modulo either child recovers its prescribed residue. Applying the binomial merge above this node recovers the degree-seven input class.

IV. ALTERNATIVE BASES AND EXACT NORMALIZATION

CRT exactness is invariant under a change of coordinates, but numerical geometry is not. Monomial coordinates $(1, z)$ in a quadratic quotient do not display the conjugate pair as an orthogonal Euclidean plane. The basis question is therefore intrinsic: find a real quotient coordinate whose multiplication law is the unit complex law.

A. The unit quotient coordinate

For $0 < \theta < \pi$, let

$$q_\theta = z^2 - 2\cos\theta z + 1, \quad e_\theta = \frac{z - \cos\theta}{\sin\theta} \quad \text{in } \mathbb{K}_N[z]/(q_\theta). \quad (15)$$

Lemma IV.1 (Quadratic unit). *The normalized generator satisfies $e_\theta^2 = -1$.*

Proof. Writing $c = \cos\theta$ and $s = \sin\theta$, the quotient relation gives $z^2 = 2cz - 1$. Therefore

$$e_\theta^2 = \frac{z^2 - 2cz + c^2}{s^2} = \frac{c^2 - 1}{s^2} = -1. \quad \square$$

Proposition IV.2 (Uniqueness among affine generators). *If a real affine class $\eta = \alpha z + \beta$ satisfies $\eta^2 = -1$ in $\mathbb{R}[z]/(q_\theta)$, then*

$$\eta = \pm \frac{z - \cos\theta}{\sin\theta}. \quad (16)$$

Proof. Reducing η^2 by $z^2 = 2cz - 1$, the coefficient of z is $2\alpha(\alpha + \beta)$. The case $\alpha = 0$ has no real solution to $\eta^2 = -1$, so $\beta = -\alpha c$. The constant term is then $-\alpha^2(1 - c^2) = -\alpha^2 \sin^2\theta$, which equals -1 precisely for $\alpha = \pm 1/\sin\theta$. $\square$

Proposition IV.3 (Canonical complex chart). *Equip the quotient with the oriented inner product*

$$\langle a + be_\theta, a' + b'e_\theta \rangle_\theta = aa' + bb'. \quad (17)$$

The map

$$\phi_\theta : a + be_\theta \mapsto a + ib \quad (18)$$

is an isometric real-algebra isomorphism from $\mathbb{K}_N[z]/(q_\theta)$, after scalar extension to $\mathbb{R}$, onto $\mathbb{C}$. At the root $z = e^{-i\theta}$, evaluation is $a + be_\theta \mapsto a - ib$.

Proof. The relation $e_\theta^2 = -1$ proves multiplicativity and bijectivity. The coefficient norm $a^2 + b^2$ is the complex modulus squared. Finally, $(e^{-i\theta} - \cos\theta)/\sin\theta = -i$. $\square$

For comparison, a raw residue $u + vz$ has normalized coordinates $(A, B) = (u + v\cos\theta, v\sin\theta)$. The coordinate matrix

$$J_\theta = \begin{bmatrix} 1 & \cos\theta \\ 0 & \sin\theta \end{bmatrix} \quad (19)$$

has singular values $\sqrt{1 \pm |\cos\theta|}$ and therefore

$$\kappa_2(J_\theta) = \sqrt{\frac{1 + |\cos\theta|}{1 - |\cos\theta|}}. \quad (20)$$

For $0 < \theta \leq \pi/2$ this is $\cot(\theta/2)$. The raw monomial chart thus becomes singular as conjugate roots approach one another, whereas the unit chart has condition number one by construction.

The choice of $e^{-i\theta}$ fixes the orientation; choosing its conjugate reverses the sign of the second coordinate.

B. The normalized four-real cell

The normalized merge of two conjugate-pair states can now be written entirely over $\mathbb{R}$. Set

$$\begin{bmatrix} r \\ i \end{bmatrix} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} o_a \\ o_b \end{bmatrix}, \quad c^2 + s^2 = 1, \quad (21)$$

and define

$$T_\theta \begin{bmatrix} e_a \\ e_b \\ o_a \\ o_b \end{bmatrix} = \begin{bmatrix} e_a + r \\ e_b + i \\ e_a - r \\ i - e_b \end{bmatrix}. \quad (22)$$

Theorem IV.4 (Local scaled orthogonality). *For every real θ ,*

$$T_\theta^\top T_\theta = 2I_4, \quad T_\theta^{-1} = \frac{1}{2} T_\theta^\top. \quad (23)$$

The binomial map $H_2(a, b) = (a + b, a - b)$ obeys the same identity.

Proof. The rotation in (21) preserves $o_a^2 + o_b^2$. Expanding the two add-subtract pairs in (22) gives

$$\|T_\theta(e_a, e_b, o_a, o_b)\|_2^2 = 2(e_a^2 + e_b^2 + r^2 + i^2) = 2\|x\|_2^2.$$

Polarization yields the matrix identity. The binomial case is the ordinary Hadamard identity. $\square$

Proposition IV.5 (Pair-merge interpretation). *Identify a real pair (a, b) with $a - ib$. Then (22), up to the displayed orientation of the second high coordinate, is the real-coordinate form of*

$$(E, O) \mapsto (E + e^{-i\theta}O, E - e^{-i\theta}O). \quad (24)$$

Thus the cell is an exact coordinate expression of a Fourier pair merge.

Proof. Multiplication of $O = o_a - io_b$ by $e^{-i\theta} = c - is$ gives $(co_a - so_b) - i(so_a + co_b) = r - ii$. Addition and subtraction from $E = e_a - ie_b$ give the four entries in (22), with the high pair stored in the declared conjugate orientation. $\square$

C. What the alternative-basis inquiry established

Two distinct alternative-basis questions sharpened this conclusion. First, balanced partitions of conjugate roots were compared by the sparsity of their CRT reduction projectors. Circularly adjacent partitions can be dense, while the covering-compatible partition in (4) retains binomial and trinomial descendants. This observation motivates the covering-map family used here; it is not a classification of all sparse CRT trees. The June 14–16 radix-four study, conducted after the first DIT construction, expressed quartic residues in Chebyshev even and odd coordinates. An isolated quartic node used fewer distinct scalar products, but its state was not the uniform quadratic two-plane used by neighboring levels. The normalized basis (15) is the exact point at which algebraic closure, unit metric, and a uniform inverse coincide.

Global orthogonality will follow after the leaf coordinate map has been defined and identified with Fourier evaluation. This order avoids assigning a uniform depth to the binomial spine: different dyadic frequency shells leave that spine at different levels.

TABLE II
COORDINATES IN A TERMINAL REAL QUADRATIC QUOTIENT

Basis	Defining relation	Evaluation at $e^{-i\theta}$
$(1, z)$	$z^2 = 2cz - 1$	$a + b(c - is)$
$(1, e_\theta)$	$e_\theta^2 = -1$	$a - ib$

V. EXACT FOURIER IDENTITY AND NON-APPROXIMATION

Orthogonality identifies the metric of B_N , but not its frequencies. Equality with the DFT follows from evaluation, independently of the round-trip theorem.

Definition V.1 (Packed real DFT). For

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i kn/N}, \quad (25)$$

define $\mathcal{D}_N^{\mathbb{R}} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ by

$$\mathcal{D}_N^{\mathbb{R}} x = (X_0, \sqrt{2}\Re X_1, -\sqrt{2}\Im X_1, \dots, \sqrt{2}\Re X_{N/2-1}, -\sqrt{2}\Im X_{N/2-1}, X_{N/2})^\top. \quad (26)$$

Definition V.2 (Normalized Bruun map). Apply $\mathcal{F}_N$ in (8). Express each quadratic residue uniquely as $a_k + b_k e_{2\pi k/N}$, retain the scalar residues at $z = 1, -1$, multiply every quadratic pair (a_k, b_k) by $\sqrt{2}$, and list these real coordinates in native factor-tree order. The resulting real linear map is

$$B_N : \mathbb{R}^N \longrightarrow \mathbb{R}^N. \quad (27)$$

The terminal pair weight is part of B_N .

Lemma V.3 (Normalized reduction diagram). *Let $\mathcal{A}_t$ be the product of quotient algebras at a cut through the Bruun tree, retaining completed scalar and quadratic leaves, and let $\rho_t : \mathcal{A}_t \rightarrow \mathcal{A}_{t+1}$ be the simultaneous remainder map at the active nodes. Let C_t list active polynomial coefficients, completed scalars, and completed quadratic coefficients in the ordered basis $(1, e_\theta)$. Define*

$$S_t = C_{t+1} \rho_t C_t^{-1}. \quad (28)$$

Then the diagram $C_{t+1} \rho_t = S_t C_t$ commutes and

$$B_N = W_N C_T \rho_{T-1} \cdots \rho_0 C_0^{-1}, \quad (29)$$

where W_N is $\sqrt{2}I_2$ on each completed quadratic pair and identity on the scalar leaves. In a pair-oriented cut schedule, every generic block of S_t is the cell T_θ , every ridge block is H_2 , and completed leaves are carry blocks.

Proof. The commuting identity is the definition of a change of coordinates on the exact remainder map. Multiplying the identities telescopes to the complete CRT reduction, and $W_N C_T$ is exactly Definition 27. For a binomial ridge, coefficient reduction is the add-subtract map H_2 . For a conjugate pair, evaluation of the two child representatives gives (24); that equation gives the displayed real cell. A factor already terminal at a cut is unchanged and therefore carried. The terminal W_N supplies the declared pair packing. $\square$

Lemma V.4 (Evaluation commutes with reduction). *If $q(\omega) = 0$ and $f \equiv r \pmod{q}$, then $f(\omega) = r(\omega)$. Consequently, the*

value at a terminal root is unchanged by every remainder map on the path from the root polynomial to its leaf.

Proof. Write $f - r = qh$ and evaluate at ω . The path statement follows by composition. $\square$

Let P_N be the signed permutation matrix that sends native factor-tree order to the order in (26). Its address law is made explicit in Section VII; all scaling is already part of B_N , and $P_N^\top P_N = I$.

Theorem V.5 (Exact Fourier identity). *For every $N = 2^L$ and every $x \in \mathbb{R}^N$,*

$$\boxed{P_N B_N x = \mathcal{D}_N^{\mathbb{R}} x}. \quad (30)$$

The equality holds as a linear identity over $\mathbb{R}$. Its matrix coefficients lie in the algebraic field $\mathbb{K}_N$ of (3).

Proof. Associate x with the polynomial in (1). The forward tree computes its exact residue at every terminal factor. The scalar leaves $z - 1$ and $z + 1$ evaluate to X_0 and $X_{N/2}$. At q_k , write the normalized residue as $a_k + b_k e_{2\pi k/N}$. By the preceding lemma and the normalized chart proposition,

$$X_k = x(e^{-2\pi i k/N}) = a_k - ib_k.$$

Conjugacy gives $X_{N-k} = \overline{X_k}$. The terminal pair scaling and P_N place $(\sqrt{2}a_k, \sqrt{2}b_k)$ in exactly the coordinates of (26). Every packed coordinate therefore agrees. $\square$

Corollary V.6 (Global normalization, inverse, and Parseval identities).

$$\begin{aligned} B_N^\top B_N &= NI, \\ (\mathcal{D}_N^{\mathbb{R}})^{-1} &= \frac{1}{N} (\mathcal{D}_N^{\mathbb{R}})^\top, \\ \|\mathcal{D}_N^{\mathbb{R}} x\|_2^2 &= N \|x\|_2^2. \end{aligned} \quad (31)$$

Thus $B_N/\sqrt{N}$ is orthogonal, every singular value is $\sqrt{N}$, and the condition number is one. In a normalized cell schedule, the local inverse halves and boundary weights distribute the inverse measure; no separate terminal normalization pass is required.

Corollary V.7 (Fourier modes and convolution). *Real cosine and sine modes map to their corresponding single conjugate-pair leaf, while the constant and alternating modes map to the scalar leaves. For circular convolution, multiplication of the complex numbers represented by the terminal pairs gives the ordinary convolution theorem after removing the boundary $\sqrt{2}$ weight from each input pair and restoring it on output.*

Equation (30) is the relevant non-approximation statement. B_N is an exact factorization of the DFT linear map. A finite-precision program evaluates this factorization with rounded coefficients and arithmetic; those residuals do not define the transform.

VI. BLOCK-ISOCLINIC FORM

The inquiry into simultaneous arithmetic across several cells has a purely linear-algebraic consequence. Cells belonging to one factor node share the same angle, so a block is a repeated rotation on mutually orthogonal two-planes.

Historically, the June 10–11 block/SIMD inquiry produced exactly the invariant proved below: equal-angle lanes carry an $O(w)$ -equivariant representation of one isoclinic rotation. The statement is independent of any instruction set.

Let $R_\theta = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and let w be the number of pairs in a node. After a fixed coordinate shuffle, the rotation portion is

$$\mathcal{R}_{\theta,w} = I_w \otimes R_\theta. \quad (32)$$

The add-subtract portion is a signed direct sum of H_2 maps.

Proposition VI.1 (Lane equivariance). *For every $U \in O(w)$,*

$$(U \otimes I_2) \mathcal{R}_{\theta,w} = \mathcal{R}_{\theta,w} (U \otimes I_2). \quad (33)$$

Moreover, the complete block is $\sqrt{2}$ times an orthogonal map.

Proof. The mixed-product rule for Kronecker products gives both sides as $U \otimes R_\theta$. The Gram identity follows from $R_\theta^\top R_\theta = I_2$ and the Hadamard identity. $\square$

Thus a collection of lanes is a representation of one algebraic operator, not a different Fourier transform. The important invariant is the common two-plane angle. DIF associates that angle with a factor-tree node, DIT with a spectral position, and DIP with a diagonal phase span.

VII. ADMISSIBLE SPECTRAL ORDERS

The native order of the terminal residues is determined by traversal of the factor tree. Its apparent shuffle becomes a precise ordering problem once locality is defined algebraically.

Definition VII.1 (Subtree-contiguous order). *A total order on the leaves of a rooted binary factor tree is subtree-contiguous if the leaves below every node form one interval of the order.*

Theorem VII.2 (Admissible orders). *An order is subtree-contiguous if and only if it is obtained by independently choosing the left-right orientation at every internal node and then listing the leaves recursively.*

Proof. Every recursive orientation clearly produces intervals. Conversely, the two child leaf sets of the root are disjoint intervals whose union is the complete order, so one precedes the other. Restriction to each child is again subtree-contiguous. Induction gives an orientation at every internal node. $\square$

This theorem explains the sorting experiments without referring to a storage device. Reordering siblings preserves the same local factor operations; interleaving leaves from unrelated subtrees fragments at least one factor interval. The admissible freedom is exactly a word of sibling reversals.

A. Frequency labels

The label of a quadratic leaf is an unordered conjugate pair $\{k, N - k\}$. The binomial spine supplies a dyadic shell before the half-angle word begins. Write

$$k = 2^v u, \quad u \text{ odd}, \quad 0 \leq v \leq L - 2. \quad (34)$$

The shell v leaves the spine through $z^{N/2^{v+1}} + 1$. Within that factor, start at $\alpha_0 = \pi/2$ and apply $\ell = L - v - 2$ half-angle decisions

$$\alpha \mapsto \frac{\alpha}{2}, \quad \alpha \mapsto \pi - \frac{\alpha}{2}. \quad (35)$$

TABLE III
NATIVE LEFT-FIRST FACTOR-TREE TRAVERSAL FOR $N = 16$. A
QUADRATIC LEAF q_k MAPS TO PACKED COORDINATES $(2k - 1, 2k)$.

Leaf	Shell v	Factor	Frequency
0	—	$z - 1$	0
1	—	$z + 1$	8
2	2	q_4	4
3	1	q_2	2
4	1	q_6	6
5	0	q_1	1
6	0	q_7	7
7	0	q_3	3
8	0	q_5	5

After ℓ decisions, $k = N\alpha/(2\pi)$ has dyadic valuation v . Choosing the representative $1 \leq k < N/2$ and adjoining 0 and $N/2$ defines a bijection from native leaves to standard packed positions.

Proposition VII.3 (Address bijection). *The shell index v together with the recursive law (35) visits every conjugate pair $\{k, N - k\}$ exactly once.*

Proof. Every $1 \leq k < N/2$ has a unique dyadic valuation v . The roots of $z^{N/2^{v+1}} + 1$ are precisely the frequencies $2^v u$ with u odd. By the half-angle factorization, the two child root sets are disjoint and their union is the parent root set. The ℓ decisions terminate in every conjugate pair of that shell once. Unique valuation separates the shells. $\square$

The address algebra is therefore an exact boundary permutation. It does not alter the factorization, its inverse, or its metric.

VIII. THE DECIMATION-IN-TIME DUAL

The Bruun remainder cascade descends from a time polynomial to spectral residues and is naturally a decimation-in-frequency (DIF) walk. An explicit decimation-in-time (DIT) dual reverses the dependency order: bit-reversed time atoms are merged into successively longer local spectra until the ordinary DFT appears in natural order.

Let rev_L reverse L binary digits and initialize

$$Y_0[j] = x_{\text{rev}_L(j)}, \quad 0 \leq j < N. \quad (36)$$

At rank s , put $e = 2^s$. For every block origin $b \equiv 0 \pmod{2e}$ and $0 \leq j < e$, define

$$U = Y_s[b + j], \quad V = Y_s[b + j + e], \quad (37)$$

$$W = e^{-2\pi i j/(2e)} V,$$

$$Y_{s+1}[b + j] = U + W, \quad (38)$$

$$Y_{s+1}[b + j + e] = U - W. \quad (39)$$

Theorem VIII.1 (Constructive DIT identity). *After rank s , each consecutive block of 2^{s+1} entries contains the natural-order DFT of the corresponding bit-reversed time comb. In particular,*

$$Y_L[k] = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N} = X_k. \quad (40)$$

Proof. For $s = 0$, (39) forms the length-two DFTs of adjacent seeds. Assume the claim for length e . The even and odd local spectra of a length $2e$ comb are the two child spectra U and V ; the shift between their time supports contributes $e^{-2\pi i j/(2e)}$. Equation (39) is therefore the exact length- $2e$ Fourier merge. Induction reaches the complete time set, and the initial bit reversal places its bits in the required merge order. $\square$

For real input, each local spectrum has conjugate symmetry. Store a conjugate pair as $(a, b) = (\Re X, -\Im X)$. Multiplication by the unit phase in (39) becomes the real rotation R_θ , and the two outputs become the normalized cell (22); DC and local Nyquist coordinates use H_2 . The real folded DIT is consequently a schedule of the same exact pair merges used by the normalized Bruun map.

Corollary VIII.2 (DIT inverse and metric). *Reversing the ranks, applying $T_\theta^\top/2$ and $H_2^\top/2$ to their local coordinates, and undoing (36) gives the two-sided inverse. After the terminal $\sqrt{2}$ pair packing, the real DIT boundary map is $\mathcal{D}_N^\mathbb{R}$ and has Gram matrix NI .*

More generally, if intermediate layouts are J_s , the conjugated ranks $\tilde{S}_s = J_{s+1} S_s J_s^{-1}$ telescope to $J_L(S_{L-1} \cdots S_0) J_0^{-1}$. This identity explains why an internal sort changes only the declared boundary layouts; equations (36)–(39) supply the specific DIT construction.

IX. AN ODD-FREQUENCY TRANSFORM IN REAL TWO-PLANE ARITHMETIC

Replacing the root polynomial $z^N - 1$ by $z^N + 1$ moves evaluation from integer frequencies to half bins. This removes the two scalar endpoints and leaves only conjugate quadratic factors.

The exact half-bin coefficients lie in $\mathbb{K}_{2N}$; all maps below are stated over $\mathbb{R}$.

Definition IX.1 (Odd-frequency DFT). For real $x \in \mathbb{R}^N$, define

$$H_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i (k+1/2)n/N}, \quad 0 \leq k < N. \quad (41)$$

Since $H_{N-1-k} = \overline{H_k}$, define the real packing

$$\mathcal{O}_N x = \sqrt{2}(\Re H_0, -\Im H_0, \dots, \Re H_{N/2-1}, -\Im H_{N/2-1})^\top. \quad (42)$$

The terminal factors of $z^N + 1$ are

$$z^2 - 2 \cos\left(\frac{2\pi(k+1/2)}{N}\right) z + 1, \quad 0 \leq k < N/2. \quad (43)$$

Each therefore possesses the same normalized generator as (15).

Theorem IX.2 (Exact ODFT). *Repeated real two-plane reduction through the factors of $z^N + 1$, followed by normalized leaf evaluation, equals (42). Moreover,*

$$\mathcal{O}_N^\top \mathcal{O}_N = NI, \quad \mathcal{O}_N^{-1} = \frac{1}{N} \mathcal{O}_N^\top. \quad (44)$$

Proof. The evaluation proof is identical to that of (30), using the roots of $z^N + 1$. Parseval for the full shifted DFT gives

$\sum_k |H_k|^2 = N\|x\|^2$. Conjugacy and the $\sqrt{2}$ packing give (44). $\square$

The recursion closes without complex scalar arithmetic. For even $N = 2M$, split x into its even and odd subsequences. With the child index reduced modulo M , one has

$$H_k^{(N)} = H_{k \bmod M}^{(M,e)} + R_{2\pi(k+1/2)/N} H_{k \bmod M}^{(M,o)}, \quad (45)$$

where every quantity on the right is unfolded to the oriented real pair $(\Re, -\Im)$, and R_θ is the real matrix in (21). For $k \geq M$, child conjugacy selects the paired index before the merge. Radix-four grouping uses the three shared rotations $R_\theta, R_{2\theta}, R_{3\theta}$ and the real quarter-turn $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Example IX.3 ($N = 4$). The positive half-bin angles are $\pi/4$ and $3\pi/4$. The only constants are $\cos(\pi/4) = \sin(\pi/4) = 1/\sqrt{2}$ and their signs. Real rotations, additions, and the quarter-turn produce the four real packed coordinates exactly. The codomain remains canonically identified with two complex spectral values, but the interior arithmetic is entirely over $\mathbb{R}^2$.

X. THE DIAGONAL PHASE TRANSFORM

DIF descends a factor tree and DIT ascends time combs. A diagonal-in-packets (DIP) transform follows a sheared coordinate frame on the finite time-frequency lattice. The construction begins with a finite Zak state, so its equality with the DFT follows by induction.

Let $\mathcal{D}_N : \mathbb{C}^N \rightarrow \mathbb{C}^N$ denote the complex map in (25), with the same negative-exponential convention.

For $0 \leq t \leq L$, put $e_t = 2^t$ and $q_t = N/e_t$, and define

$$Z_t[d, j] = \sum_{r=0}^{e_t-1} x_{j+rq_t} e^{-2\pi i d r / e_t}, \quad (46)$$

$$0 \leq d < e_t, \quad 0 \leq j < q_t.$$

At $t = 0$, $Z_0[0, j] = x_j$; at $t = L$, $Z_L[d, 0] = X_d$.

Lemma X.1 (Explicit Zak advance). *For $0 \leq d < 2e_t$ and $0 \leq j < q_t/2$,*

$$Z_{t+1}[d, j] = Z_t[d \bmod e_t, j] + e^{-2\pi i d / (2e_t)} Z_t[d \bmod e_t, j + q_t/2]. \quad (47)$$

Thus every advance is two-sparse.

Proof. Split the sum defining Z_{t+1} into even and odd comb indices. The even indices give the first child state. Writing an odd index as $2r+1$ contributes the phase $e^{-2\pi i d / (2e_t)}$ and the offset $j + q_t/2$, giving the second state in (47). $\square$

A. Admissible monomial shears

For an integer σ_t , impose

$$\sigma_t q_t^2 \equiv 0 \pmod{N}, \quad c_t = \frac{\sigma_t q_t^2}{N} \in \mathbb{Z}. \quad (48)$$

The sheared Zak frame is the monomial unitary

$$(\mathcal{S}_t^{\sigma_t} Z_t)[\kappa, j] = e^{i\pi \sigma_t j^2 / N} Z_t[(\kappa - \sigma_t j - c_t 2^{t-1}) \bmod e_t, j]. \quad (49)$$

At $t = 0$ the final term is interpreted as zero. Admissibility makes the row shift integral on the finite lattice. The phase factor makes $\mathcal{S}_t^{\sigma_t}$ unitary rather than merely permutational.

Let A_t denote (47) and define the DIP stage

$$A_t^{\text{DIP}} = \mathcal{S}_{t+1}^{\sigma_{t+1}} A_t (\mathcal{S}_t^{\sigma_t})^{-1}. \quad (50)$$

Theorem X.2 (DIP exactness). *For any admissible schedule with $\mathcal{S}_0^{\sigma_0} = \mathcal{S}_L^{\sigma_L} = I$,*

$$A_{L-1}^{\text{DIP}} \cdots A_0^{\text{DIP}} = A_{L-1} \cdots A_0 = \mathcal{D}_N. \quad (51)$$

Every stage remains a two-sparse packet merge. In the centered diagonal coordinate

$$\delta = (\kappa - \sigma_{t+1}j - c_{t+1}e_t) \bmod 2e_t, \quad (52)$$

its butterfly phase is $e^{-2\pi i \delta / (2e_t)}$, a function of one diagonal coordinate.

Proof. The first equality follows by cancellation of adjacent monomial unitaries. The Zak boundary identities and (47) give the second. Substitution of (49) into (50), using (48), leaves two input coordinates and collects their relative phase into (52). Hence the conjugated stage is still two-sparse and its rotation is constant along each diagonal. $\square$

A canonical complex schedule is

$$\sigma_0 = \sigma_L = 0, \quad \sigma_t = 2^{\max(0, 2t-L)}, \quad 1 \leq t < L, \quad (53)$$

which satisfies (48). If packets are phase-referenced at their centers, the quadratic factor in (49) is absorbed into the packet definition. Diagonal-major storage then has the same arithmetic recurrence as (47); the diagonal motion resides in the physical meaning of (δ, j) .

B. Real folding and the span form

For real x , $Z_t[e_t - d, j] = \overline{Z_t[d, j]}$. Folding each conjugate pair into $(a, b) = (\Re, -\Im)$ changes the phase in (47) into the real rotation of (22). Conjugation remains span-local precisely when

$$2\sigma_t \equiv 0 \pmod{e_t}; \quad (54)$$

for $e_t \geq 2$, the real-fold-compatible shears modulo e_t are therefore 0 and $e_t/2$. At $e_0 = 1$, only the zero class exists. The half-turn exchanges sibling orientation without changing the covered frequencies.

In the equivalent depth-first span notation, e is a power of two dividing N , $0 < d < e/2$, and a generic span (d, e) uses angle $\theta = \pi d/e$. Its two children are $(d, 2e)$ and $(e-d, 2e)$. At terminal denominator E , its leaves form

$$\Lambda(d, E) = \{jE + d, jE - d : j \in \mathbb{Z}\} \pmod{N}. \quad (55)$$

Doubling lifts a congruence class into exactly the two child classes, so induction proves that sibling combs are disjoint up to conjugacy and partition the parent comb. The ridge cases $d = 0$ and $d = e/2$ advance by a Hadamard map and promote the new quarter-frequency pair; they are excluded from the generic two-child rule.

Proposition X.3 (Span-Zak equivalence). *Under centered packet coordinates, a generic span (d, e) is the folded Zak*

pair $(Z_t[d, \cdot], Z_t[e - d, \cdot])$ with $e = 2^t$. Its normalized span cell is exactly the real folding of (47), and its children have labels $(d, 2e)$ and $(e - d, 2e)$.

Proof. The phase in (47) is $e^{-2\pi id/(2e)} = e^{-i\pi d/e}$. In the oriented pair $(\mathbb{R}, -\mathbb{S})$ it is the rotation $R_{\pi d/e}$. Adding and subtracting the two comb halves gives $T_{\pi d/e}$, while conjugacy sends the second output to row $e - d$ at denominator $2e$. Induction from the ridge initialization proves the identification at every span. $\square$

This span recursion is the in-place real representative of the diagonal packet walk.

XI. A PHYSICALLY REALIZABLE FOURIER TRANSFORM

Here “physically realizable” has a precise ideal linear meaning. A signed state is represented by two nonnegative rails; local transport matrices have nonnegative entries and preserve a declared mass or constant potential; and an observable is recovered by rail difference. A second realization uses convex equilibrium junctions with an explicit quadratic energy. These element laws are sufficient to realize every normalized Bruun stage.

A. Positive-cone lift

For a real matrix $M \in \mathbb{R}^{m \times n}$, define $M^+ = \max(M, 0)$ and $M^- = \max(-M, 0)$ entrywise, and set

$$C(M) = \begin{bmatrix} M^+ & M^- \\ M^- & M^+ \end{bmatrix}, \quad \Pi_n = \begin{bmatrix} I_n & -I_n \end{bmatrix}. \quad (56)$$

Theorem XI.1 (Exact cone projection). *For every real matrix M ,*

$$\Pi_m C(M) = M \Pi_n. \quad (57)$$

Consequently, for any composable sequence $M_1, \dots, M_s$,

$$\Pi C(M_s) \cdots C(M_1) = M_s \cdots M_1 \Pi. \quad (58)$$

Proof. Since $M = M^+ - M^-$,

$$\Pi_m C(M) = [M^+ - M^-, M^- - M^+] = M \Pi_n.$$

Repeated substitution proves (58). $\square$

When dimensions are clear, subscripts on Π are suppressed. The cone lift maps $\mathbb{R}_+^{2n}$ to $\mathbb{R}_+^{2m}$ and stores additional common mode in $\ker \Pi_n = \{(u, u) : u \in \mathbb{R}^n\}$. Equal amounts may be removed from both rails without changing the signed observable. Composition is exact after projection even though the lifted matrices need not multiply as $C(AB)$.

Define rail canonicalization componentwise by

$$K(p, n) = (p - r, n - r), \quad r_i = \min(p_i, n_i), \quad (59)$$

and signed injection by $J(x) = (x^+, x^-)$. Canonicalization gives a literal fixed-point law for the nonnegative state.

Theorem XI.2 (Canonical rail fixed points). *The map K is idempotent, $\Pi K = \Pi$, and*

$$\text{Fix}(K) = \{(p, n) : p_i n_i = 0 \text{ for every } i\}. \quad (60)$$

For every real M and signed x ,

$$KC(M)J(x) = J(Mx). \quad (61)$$

Consequently, inserting K after every lifted Fourier stage yields the canonical injection of the exact signed stage product.

Proof. After subtracting r_i , at least one rail in each pair is zero; a second application subtracts zero, proving idempotence and (60). Equal subtraction preserves rail difference. Equation (57) shows that $C(M)J(x)$ has signed projection Mx ; canonicalization selects the unique disjoint-support representative of that projection, namely $J(Mx)$. Induction proves the stage product statement. $\square$

B. Conservative mass gauge

Consider a factorization $x_{s+1} = M_s x_s$. Fix $g_S = \mathbf{1}$ and recurse backward by

$$g_s = |M_s|^\top g_{s+1}, \quad A_s = \text{diag}(g_{s+1}) M_s \text{diag}(g_s)^{-1}. \quad (62)$$

Theorem XI.3 (Column-stochastic cone stages). *If every component of g_s is positive, then $C(A_s)$ is nonnegative and column-stochastic. The diagonal gauges telescope, and terminal signed projection with the declared boundary calibration recovers $M_{S-1} \cdots M_0$ exactly.*

Proof. For column j ,

$$\sum_i |(A_s)_{ij}| = \frac{\sum_i (g_{s+1})_i |(M_s)_{ij}|}{(g_s)_j} = 1.$$

Each column of $C(A_s)$ therefore sums to one. With $D_s = \text{diag}(g_s)$, multiplying the gauged stages gives

$$A_{S-1} \cdots A_0 = D_S (M_{S-1} \cdots M_0) D_0^{-1}. \quad (63)$$

Injecting $J(D_0 x)$ and decoding by $D_S^{-1} \Pi$ therefore recovers the original signed product. Here $g_S = \mathbf{1}$, so $D_S = I$. $\square$

This construction preserves total two-rail mass at every transport stage. Annihilation of equal rail amounts is a separately ledgered conversion into a null observable; it cannot change any Fourier coordinate.

C. Kinematic and equilibrium gauge

For a nonnegative cone stage C_s with no zero row, choose $d_0 > 0$ and define componentwise

$$d_{s+1} = (C_s(1/d_s))^{-1}, \quad B_s = \text{diag}(d_{s+1}) C_s \text{diag}(d_s)^{-1}. \quad (64)$$

Every normalized Bruun cell has a nonzero entry in each row, so direct sums and coordinate permutations of its cone lifts satisfy the hypothesis.

Proposition XI.4 (Convex junction form). *$B_s \mathbf{1} = \mathbf{1}$. Hence each output coordinate is a convex combination of the input coordinates and can be decomposed into binary interpolation junctions.*

Proof. Substitution of (64) gives $B_s \mathbf{1} = \text{diag}(d_{s+1}) C_s(1/d_s) = \mathbf{1}$. $\square$

Writing $\Delta_s = \text{diag}(d_s)$ gives the boundary-calibrated product

$$B_{S-1} \cdots B_0 = \Delta_S C_{S-1} \cdots C_0 \Delta_0^{-1}. \quad (65)$$

Thus a cone input is encoded by Δ_0 , and the output is decoded by Δ_S^{-1} before applying Π .

Theorem XI.5 (Exact clamped equilibrium). *Let $B \geq 0$ be row-stochastic and hold x fixed. The energy*

$$E_x(y) = \frac{1}{2} \sum_{j,i} B_{ji} (y_j - x_i)^2 \quad (66)$$

has the unique equilibrium

$$y = Bx. \quad (67)$$

Proof. Row stochasticity gives $\partial E_x / \partial y_j = y_j - \sum_i B_{ji} x_i$. The Hessian with respect to y is I , so the stationary point (67) is unique. $\square$

The theorem applies to an isolated stage with prescribed input, or once to the row-stochastic product of all kinematic stages. The latter makes the complete Fourier boundary response an exact static equilibrium. The sparse exact realization is instead the ideal kinematic-constraint or reciprocal-force cascade. A reciprocal linkage obeying $q_{\text{in}} = A^\top q_{\text{out}}$ also transmits generalized force as $f_{\text{out}} = A f_{\text{in}}$ by equality of virtual work. Thus the transpose kinematic constraint and forward Fourier force map are the same reciprocal linear element.

D. Field-phase DIP form

Before conjugate folding, the normalized complex butterfly is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & w \\ 1 & -w \end{bmatrix}, \quad |w| = 1. \quad (68)$$

It is a fixed phase delay followed by a lossless balanced coupler, since the matrix in (68) is unitary. In the real two-plane description, the phase delay is R_θ and addition is a linear junction. Conjugation required by a folded half-spectrum is a separate real reflection. Thus the unfurled DIP lattice has an exact passive scattering realization of $N^{-1/2} \mathcal{D}_N$; a declared boundary gain $\sqrt{N}$ recovers the unnormalized Fourier convention. The folded construction adds the declared reflection element. The cone and field constructions are two ideal physical categories for the same Fourier observable.

Corollary XI.6 (Normalized Bruun realization). *Apply the cone lift and either diagonal gauge to any exact stage factorization of B_N . With the boundary encodings in (63) or (65), rail-difference projection equals $B_N x$, and standard output permutation gives $\mathcal{D}_N^{\mathbb{R}} x$ exactly.*

XII. EXACT EXAMPLES AND FINITE-PRECISION AUDIT

The examples first display exact identities. The numerical tables then audit binary64 evaluations of the already defined operators.

TABLE IV
BINARY64 DIAGONAL REAL-TRANSFORM AUDIT: MAXIMUM ABSOLUTE PACKED-BIN ERROR FOR PCG64 SEED 7 STANDARD-NORMAL INPUTS.

N	$\ \hat{B}_N x - \mathcal{D}_N^{\mathbb{R}} x\ _\infty$
16	6.66×10^{-16}
64	3.66×10^{-15}
512	1.57×10^{-14}
4096	7.11×10^{-14}

A. The exact $N = 4$ transform

With the packing of (26),

$$\mathcal{D}_4^{\mathbb{R}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ \sqrt{2} & 0 & -\sqrt{2} & 0 \\ 0 & \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -1 & 1 & -1 \end{bmatrix}. \quad (69)$$

Direct multiplication gives $(\mathcal{D}_4^{\mathbb{R}})^\top \mathcal{D}_4^{\mathbb{R}} = 4I$. Its middle two rows are $\sqrt{2}$ times the normalized residue coordinates of $x(z)$ modulo $z^2 + 1$; its first and last rows are evaluation modulo $z - 1$ and $z + 1$.

For $N = 8$, Fig. 1 adds the two quadratic factors with coefficients $\pm\sqrt{2}$. Their normalized angles are $\pi/4$ and $3\pi/4$. Every irrational coefficient is an element of $\mathbb{K}_8 = \mathbb{Q}(\sqrt{2})$, so the factorization and the CRT return are exact over that field.

An explicit radical form makes the leaf values visible. Put $d_j = x_j + x_{j+4}$, $p_j = x_j - x_{j+4}$ for $0 \leq j < 4$, and $a = 1/\sqrt{2}$. Then

$$X_0 = d_0 + d_1 + d_2 + d_3, \quad (70)$$

$$X_4 = d_0 - d_1 + d_2 - d_3, \quad (71)$$

$$X_2 = (d_0 - d_2) - i(d_1 - d_3), \quad (72)$$

$$X_1 = p_0 + a(p_1 - p_3) - i[p_2 + a(p_1 + p_3)], \quad (73)$$

$$X_3 = p_0 - a(p_1 - p_3) - i[a(p_1 + p_3) - p_2]. \quad (74)$$

The remaining bins are the conjugates of X_1, X_2, X_3 . Substitution into (25) proves these identities term by term, and the local inverse cells recover each d_j, p_j and then each x_j .

B. Fourier and diagonal audits

Table IV reports maximum absolute coordinate error against direct Fourier summation for the real diagonal walk. Inputs were successive standard-normal vectors from NumPy PCG64 with seed 7, sampled in ascending order of N . The ordered and sheared complex Zak products agreed entrywise in the binary64 evaluation, while their error against direct summation grew with the ordinary accumulation length.

Let $\hat{B}_N$ denote the binary64 evaluation of the exact normalized factorization B_N for the stated deterministic input.

The faithful monomial-coordinate CRT reduction at $N = 512$ gave a forward residual 7.0×10^{-13} and a forward-inverse residual 1.1×10^{-14} . For the normalized map, direct matrix audits over $N = 4, 8, \dots, 512$ placed orthogonality residuals between 10^{-16} and 7.5×10^{-15} , with measured condition number at most 1.0000000000000001. These measurements distinguish the algebraic identity from its floating evaluation while confirming the normalization law.

TABLE V
BINARY64 $N = 8$ PHYSICAL-IDENTITY AUDIT. EACH ROW IS THE
INFINITY NORM OF THE STATED MATRIX OR VIRTUAL-WORK IDENTITY;
GENERIC VECTORS USE PCG64 SEED 20260812.

Identity	Maximum defect
Normalized Fourier factorization	1.05×10^{-15}
Cone intertwining	0
Mass-gauge column sums	0
Kinematic-gauge row sums	1.11×10^{-16}
Reciprocal virtual work	8.88×10^{-16}
Spring equilibrium reconstruction	1.92×10^{-16}

C. Positive-cone and equilibrium audit

For a 512-point normalized factorization and a standard-normal PCG64 seed 43 input, the backward gauge produced an input potential range [256, 1324]. The maximum column-stochasticity defect was 2.22×10^{-16} , the projected forward residual was 2.16×10^{-14} , and the forward-inverse residual was 1.78×10^{-15} . A normally distributed signed input permitted 97.4% of the lifted mass to be removed as equal-rail common mode; an impulse required no annihilation.

An exact-structure $N = 8$ mechanical certificate is summarized in Table V. The binary interpolation decomposition contained 44 bars—36 midpoint and eight with ratio $\sqrt{2}-1$ —plus twelve pass-through rows. These counts describe one decomposition of the stochastic stages, not additional assumptions in the realizability theorem.

The delay-line DIP evaluation gave maximum Fourier residuals 3.97×10^{-15} , 2.04×10^{-14} , and 7.32×10^{-14} at $N = 64, 512$, and 4096, respectively. All reported errors are maximum absolute coordinate errors for deterministic test vectors or seeded pseudorandom vectors evaluated in binary64 arithmetic.

XIII. RELATION TO PRIOR WORK

Cooley and Tukey established the recursive decomposition now associated with complex-radix FFTs [2]. Bruun instead retained real conjugate factors and interpreted the recursion through real z -transform filters [1]. The construction here retains Bruun’s half-angle factor tree and supplies explicit inverse interpolation, normalized quotient coordinates, and boundary maps.

Chen, Sorensen, and Jones specialized Bruun’s structure to real-symmetric data [3]. The normalized quotient map here instead treats arbitrary real input and derives its inverse and metric from the complete real CRT.

Chinese-remainder formulations of Fourier algorithms appear in the work of Good, Winograd, and the algebraic signal-processing literature [4], [5], [6], [7]. In that language, Cooley-Tukey and Bruun are instances of a common polynomial factorization scheme expressed in different intermediate bases [7]. The present normalization makes this basis choice explicit at every real quadratic leaf and follows its metric consequence through the complete product.

Real-input FFT algorithms commonly exploit conjugate symmetry to reduce arithmetic and storage [8], [9]. The packed transform (26) uses the same spectral degrees of freedom, but

derives them directly from real quotient algebras. The ODFT is related to shifted and half-bin Fourier grids; its contribution here is closure of the same normalized two-plane arithmetic on the factors of $z^N + 1$.

Stockham schedules and DIT/DIF variants reorder intermediate Fourier states [10], [9]. The phase-packet result states the common principle as an exact conjugation of each stage, with the DIP schedule selected by a lattice shear. The positive-cone result is independent of a particular schedule: it lifts any signed stage factorization and preserves the complete product under rail-difference projection.

XIV. CONCLUSION

Bruun’s real factor tree supports a complete Fourier theory. Explicit local CRT interpolation makes the tree two-sided. The quotient generator e_θ gives every conjugate pair a unit Euclidean coordinate, turning each node into a scaled-orthogonal rotation and Hadamard map. Evaluation at the terminal roots then proves equality with the packed DFT, not merely invertibility or numerical agreement. The same exact operator admits DIF, DIT, and diagonal phase-packet walks; the factors of $z^N + 1$ give an odd-frequency companion; and positive-cone gauges give conservative transport and exact equilibrium realizations in the declared ideal linear categories. A remaining algebraic problem is to classify all sparse real CRT trees that are simultaneously scaled orthogonal and closed under a finite family of local coordinate types.

APPENDIX A

EXPANDED LOCAL CRT ALGEBRA

For completeness, the trinomial merge can be checked without invoking an abstract CRT theorem. Put $u = z^{h/2}$. Modulo p_+ ,

$$u^2 + bu + 1 = 0, \quad u^{-1} = -(u + b). \quad (75)$$

Since $p_- = p_+ - 2bu$, one has

$$p_- \frac{u + b}{2b} \equiv -u(u + b) = 1 \pmod{p_+}. \quad (76)$$

Thus the correction from r_- to r_+ is the child difference multiplied by the inverse in (76). Degree reduction modulo p_+ makes $\deg t < h$, and multiplication by p_- gives a representative of degree below $2h$. This is exactly the unique parent coordinate range.

For binomial siblings, $p_- = z^h - 1 \equiv -2 \pmod{p_+}$. The correction equation $r_- + p_- t \equiv r_+$ gives $t = (r_- - r_+)/2$. In normalized coordinates, every four-real pair merge has inverse $T_\theta^\top/2$. Across the complete unbalanced tree, these local measures combine with the terminal pair weights to give the global inverse $B_N^{-1} = B_N^\top/N$ proved from Fourier evaluation in Section V.

APPENDIX B

STAGEWISE ORTHOGONALITY

The normalized cell has the explicit matrix

$$T_\theta = \begin{bmatrix} 1 & 0 & c & -s \\ 0 & 1 & s & c \\ 1 & 0 & -c & s \\ 0 & -1 & s & c \end{bmatrix}. \quad (77)$$

TABLE VI
ALGEBRAIC ROLES OF THE TRANSFORMS DEVELOPED HERE

Object	Root polynomial	Interior state	Boundary order	Exact inverse	Metric
Normalized Bruun DIF	$z^N - 1$	real CRT residues	native leaves	local CRT ascent	$B^T B = NI$
Bruun DIT	$z^N - 1$	spectral combs	natural spectrum	reversed cells	$B^T B = NI$
DIP	$z^N - 1$	sheared Zak packets	natural spectrum	reversed shear	$B^T B = NI$
ODFT	$z^N + 1$	real half-bin pairs	conjugate pairs	local pair inverse	$O^T O = NI$
Cone lift	either	nonnegative rails	projected boundary	signed inverse after projection	gauge-dependent

Its columns have squared norm two. The first and third columns are orthogonal because their c contributions cancel; the second and fourth behave similarly. The remaining inner products vanish by either sign cancellation or $cs - sc = 0$. Hence $T_\theta^T T_\theta = 2I$ entrywise.

Direct sums of equal-rank cells retain this Gram identity, and permutations retain it by conjugation. The complete Bruun tree also contains shell carry coordinates because its binomial spine is unbalanced; those coordinates and the terminal $\sqrt{2}$ pair weights are included in Definition 27. The schedule-independent global statement is therefore obtained from the exact identity $P_N B_N = \mathcal{D}_N^{\mathbb{R}}$ and Fourier Parseval:

$$B_N^T B_N = B_N^T P_N^T P_N B_N = NI. \quad (78)$$

This argument covers every exact schedule of the same boundary map without assigning an artificial common depth to the terminal factors.

APPENDIX C ADDRESS ALGEBRA

Number the nontrivial heap leaves by $1 \leq m < N/2$. Put

$$t = \lfloor \log_2 m \rfloor, \quad r = m - 2^t, \quad 0 \leq r < 2^t. \quad (79)$$

Let rev_t reverse the t binary digits of its argument, and let gray^{-1} be inverse Gray decoding, defined by the prefix XOR recurrence $b_{t-1} = g_{t-1}$ and $b_j = b_{j+1} \text{ xor } g_j$. The standard conjugate-pair label is

$$k(m) = (2 \text{gray}^{-1}(\text{rev}_t(r)) + 1) 2^{L-2-t}. \quad (80)$$

Theorem C.1 (Heap-shell bijection). *Equation (80) is a bijection from the nontrivial Bruun leaves to $1 \leq k < N/2$. The heap shell $2^t \leq m < 2^{t+1}$ maps exactly to the frequency shell with dyadic valuation $L - 2 - t$.*

Proof. Bit reversal and inverse Gray decoding are bijections on the t -bit words. The parenthesized term in (80) therefore visits every odd integer from one through $2^{t+1} - 1$ exactly once. Multiplication by 2^{L-2-t} gives precisely one dyadic-valuation shell. The shells are disjoint and their union is $1, \dots, N/2 - 1$. $\square$

For $N = 16$, the leaf indices $m = 1, \dots, 7$ map to

$$(k(1), \dots, k(7)) = (4, 2, 6, 1, 7, 3, 5), \quad (81)$$

which is the quadratic-leaf portion of Table III.

The sibling-orientation word acts by replacing any subtree interval with its reverse. Because this action never changes the underlying quadratic factor, it changes only the output

permutation. A standard-frequency conversion may therefore be attached at the boundary of any DIF, DIT, or DIP schedule without changing an interior coefficient.

APPENDIX D POSITIVE-CONE GAUGES AND COMMON MODE

Every signed vector x admits nonnegative representatives $u = (x^+, x^-)^T$ with $\Pi u = x$. Representatives are not unique: $(x^+ + a, x^- + a)$ has the same projection for every $a \geq 0$. This freedom is the common-mode gauge.

Canonicalization in (59) removes exactly $2 \sum_i \min(p_i, n_i)$ units of two-rail mass. The identity $K^2 = K$ makes the remaining state a fixed point of further annihilation, while $\Pi K = \Pi$ makes the removed mass an exact common-mode ledger. No signed Fourier information is carried by that ledger.

For a stage M , $C(M)u$ is nonnegative and projects to Mx . For two stages,

$$\Pi C(B)C(A)u = B \Pi u, \quad (82)$$

even when $C(B)C(A) \neq C(BA)$. Their difference maps every representative into $\ker \Pi$. Hence annihilating equal rails between stages selects a smaller representative without changing any downstream signed observable.

For the backward gauge, positivity of g_s follows whenever every input coordinate influences at least one terminal coordinate. Normalized Bruun cells have this property. Column stochasticity gives

$$\mathbf{1}^T C(A_s)u = \mathbf{1}^T u, \quad (83)$$

so transported mass is conserved exactly. For the forward gauge, row stochasticity makes constant potential an exact fixed point. The two gauges are dual boundary normalizations of the same positive cone operator.

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